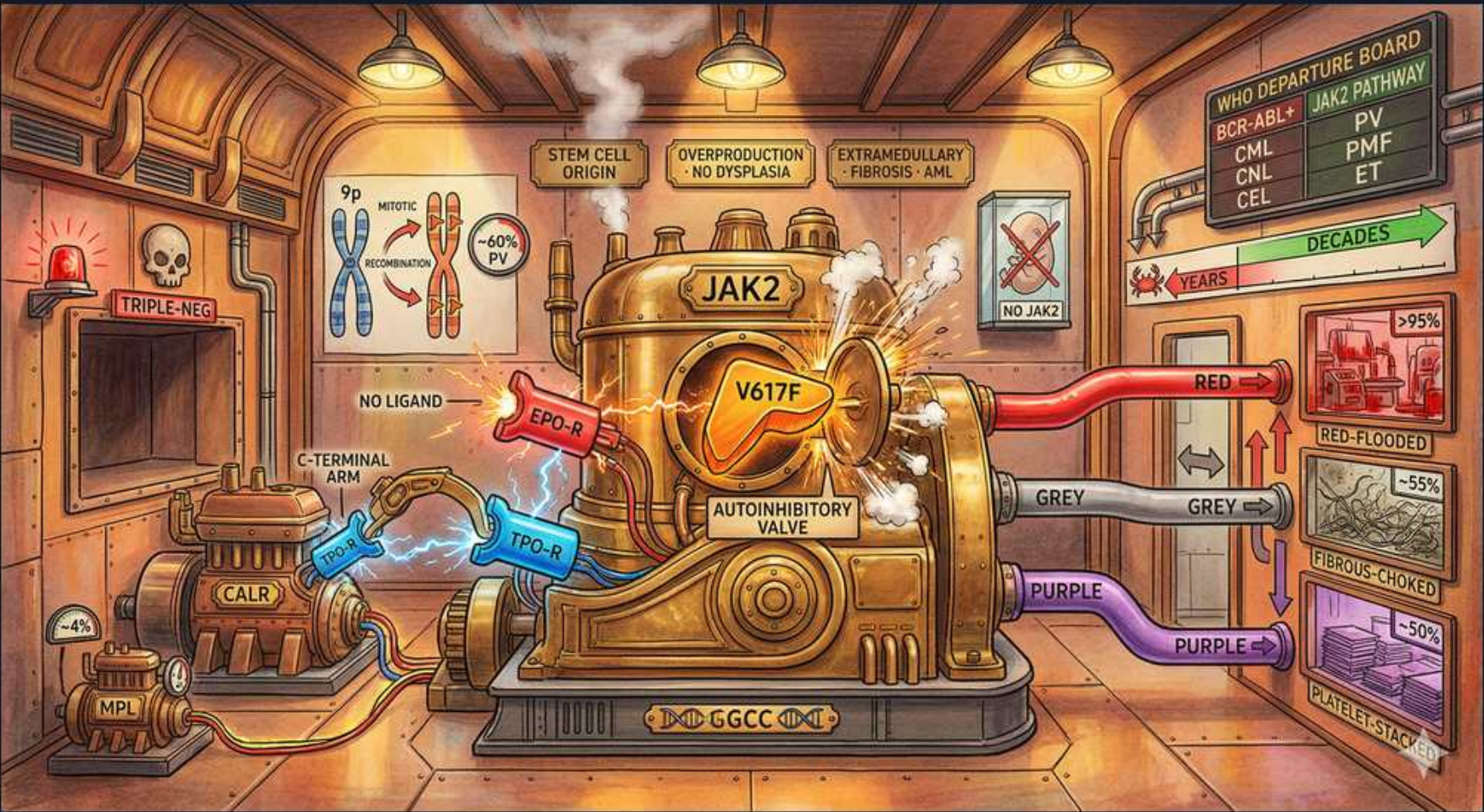


Polycythemia / MPN — Polycythemia Vera pathophysiology (JAK2)



1. JAK2 V617F engine

WHAT YOU SEE

Brass engine labeled JAK2 with glowing V617F core

WHAT IT ENCODES

JAK2 V617F (a valine-to-phenylalanine substitution at codon 617 in the JH2 pseudokinase domain) is the driver mutation in ~95% of polycythemia vera. It creates a constitutively active cytoplasmic tyrosine kinase that fires JAK-STAT signaling without ligand. The ~3-5% of JAK2 V617F-negative PV cases carry JAK2 exon-12 mutations — so essentially 100% of PV is JAK2-driven.

BOARD TRAP

WRONG: the driver here is BCR-ABL. BCR-ABL defines CML (the Philadelphia chromosome MPN); PV is BCR-ABL NEGATIVE and JAK2-mutated. Discriminator: send BCR-ABL to exclude CML, then JAK2 to confirm PV.

2. Autoinhibitory valve broken

WHAT YOU SEE

Valve labeled AUTOINHIBITORY at base of engine

WHAT IT ENCODES

Normally the JH2 pseudokinase domain auto-inhibits the JH1 kinase domain, keeping JAK2 off until a cytokine binds its receptor. V617F disrupts this autoinhibitory brake, so the kinase is permanently 'on' — ligand-independent, constitutive JAK-STAT/PI3K/MAPK signaling driving proliferation.

BOARD TRAP

WRONG: V617F is a loss-of-function mutation that disables JAK2. It is GAIN-of-function — it removes the autoinhibitory brake, producing constitutive activity, not loss of signaling.

3. EPO-R no ligand

WHAT YOU SEE

Blue EPO-R receptor with NO LIGAND label

WHAT IT ENCODES

Mutant JAK2 signals through the erythropoietin receptor (and TPO-R/G-CSF-R scaffolds) WITHOUT erythropoietin binding. Because red-cell production runs autonomously, the body suppresses EPO — so serum EPO is characteristically LOW in PV. A low EPO is the key lab discriminator separating PV from secondary polycythemia.

BOARD TRAP

WRONG: serum EPO is high in polycythemia vera. EPO is LOW in PV (autonomous, feedback-suppressed); a HIGH EPO points to SECONDARY polycythemia (hypoxia, EPO-secreting tumor, exogenous EPO).

4. >95% PV / JAK2

WHAT YOU SEE

WHO departure board, RED row >95%

WHAT IT ENCODES

WHO criteria leverage that JAK2 mutation is present in >95% of PV (V617F plus exon-12). Major criteria: elevated Hgb/Hct or red-cell mass, characteristic hypercellular trilineage marrow with pleomorphic megakaryocytes, and a JAK2 mutation; the minor criterion is subnormal serum EPO. JAK2 testing is the pivotal molecular step.

BOARD TRAP

WRONG: a negative JAK2 V617F test rules out PV. About 3-5% of PV is V617F-negative but exon-12 positive — if V617F is negative with a low EPO and high Hct, reflex to JAK2 exon-12 testing before excluding PV.

5. CALR / MPL (no JAK2)

WHAT YOU SEE

C-terminal arm CALR and MPL receptors bottom-left

WHAT IT ENCODES

CALR and MPL mutations drive the JAK2-wild-type MPNs — essential thrombocythemia (ET) and primary myelofibrosis (PMF) — by activating the same downstream JAK-STAT axis (CALR binds and activates MPL/TPO-R). They do NOT cause polycythemia vera.

BOARD TRAP

WRONG: CALR mutations can drive polycythemia vera. CALR/MPL are seen in ET and PMF only; PV is essentially always JAK2-mutated — finding CALR points you away from PV toward ET/PMF.

6. Triple-negative

WHAT YOU SEE

Skull TRIPLE-NEG tag, far left

WHAT IT ENCODES

'Triple-negative' MPN lacks detectable JAK2, CALR, and MPL mutations. It is rare, occurs in ET/PMF (not PV, which is virtually always JAK2-positive), and in PMF carries the worst prognosis. These cases often harbor non-canonical or high-molecular-risk subclonal mutations.

BOARD TRAP

WRONG: triple-negative means a reactive, non-clonal blood disorder. The disease is still clonal/neoplastic — driven by other lesions — and in PMF triple-negativity is the most aggressive subgroup.

7. RED = erythrocytosis

WHAT YOU SEE

Red pipe to RED FLOODED gauge >95%

WHAT IT ENCODES

The hallmark of PV is erythrocytosis — elevated hematocrit/hemoglobin from autonomous red-cell overproduction. Diagnostic thresholds: Hgb >16.5 g/dL (men)/>16.0 (women) or Hct >49%/>48%. The resulting hyperviscosity causes headache, plethora, dizziness, blurred vision, and thrombosis (including arterial/venous and Budd-Chiari).

BOARD TRAP

WRONG: an isolated high hematocrit always means polycythemia vera. First exclude RELATIVE polycythemia (dehydration/plasma contraction) and SECONDARY causes (hypoxia, smoking/CO, EPO-secreting tumor); confirm true PV with low EPO + JAK2.

8. GREY = WBC

WHAT YOU SEE

Grey pipe to FIBROUS/MOVED ~55% gauge

WHAT IT ENCODES

PV is a panmyelosis — leukocytosis (neutrophilia) accompanies the erythrocytosis in many patients, reflecting trilineage stem-cell expansion. Leukocytosis is itself an adverse prognostic and thrombotic risk factor in PV, though the red line dominates the phenotype.

BOARD TRAP

WRONG: leukocytosis excludes PV and indicates infection or CML. PV is a trilineage MPN, so leukocytosis is expected; the distinction from CML is BCR-ABL (positive in CML, negative in PV).

9. PURPLE = platelets

WHAT YOU SEE

Purple pipe to PLATELET gauge ~30%

WHAT IT ENCODES

Thrombocytosis is the third arm of PV's panmyelosis. Counts can rise high enough to provoke ACQUIRED von Willebrand syndrome (vWF multimers consumed by excess platelets), causing paradoxical BLEEDING at extreme counts (typically >1,000-1,500 x10^9/L) despite the prothrombotic state.

BOARD TRAP

WRONG: extreme thrombocytosis in PV means even greater clotting risk and warrants more aggressive anticoagulation. Very high platelet counts can cause acquired von Willebrand disease and BLEEDING — avoid aspirin/anticoagulation until vWF activity is checked.

10. Overproduction, no dysplasia

WHAT YOU SEE

OVERPRODUCTION · NO DYSPLASIA banner top center

WHAT IT ENCODES

MPNs (PV, ET, PMF, CML) are defined by EFFECTIVE overproduction of mature, functional blood cells WITHOUT dysplasia. This contrasts with myelodysplastic syndromes (MDS), where INEFFECTIVE, dysplastic hematopoiesis produces cytopenias despite a cellular marrow.

BOARD TRAP

WRONG: PV shows dysplastic, ineffective hematopoiesis. That describes MDS — MPNs feature effective overproduction of normal-appearing mature cells; presence of dysplasia/cytopenias should redirect you toward MDS or MDS/MPN overlap.

11. Stem cell origin

WHAT YOU SEE

STEM CELL ORIGIN tag top left

WHAT IT ENCODES

PV arises from a single transformed multipotent hematopoietic STEM cell, which is why all three lineages can be involved (panmyelosis). The clonal stem-cell origin also explains the long natural history and capacity to evolve into post-PV myelofibrosis or AML.

BOARD TRAP

WRONG: PV is a disease of mature committed erythroid progenitors only. It is a clonal STEM-cell disorder — the multipotent origin accounts for trilineage involvement and transformation potential.

12. Extramedullary fibrosis → AML

WHAT YOU SEE

EXTRAMEDULLARY FIBROSIS · AML tag top

WHAT IT ENCODES

PV can evolve to a 'spent phase' / post-PV myelofibrosis (marrow fibrosis, falling counts, splenomegaly from extramedullary hematopoiesis) and ultimately transform to secondary AML. Risk of leukemic transformation rises with disease duration and with cytoreductive exposure.

BOARD TRAP

WRONG: PV never transforms — it is a benign chronic erythrocytosis. PV is a clonal neoplasm that can progress to post-PV myelofibrosis and secondary AML; the latter is treatment-resistant with poor survival.

13. JAK2 pathway / decades

WHAT YOU SEE

YEARS→DECADES timeline on board

WHAT IT ENCODES

Ph-negative MPNs are chronic, indolent diseases evolving over years to decades. Management of PV aims to prevent thrombosis (the leading cause of morbidity/mortality): therapeutic phlebotomy to Hct <45%, low-dose aspirin, and cytoreduction (hydroxyurea first-line; ruxolitinib or interferon second-line) in high-risk patients (age >60 or prior thrombosis).

BOARD TRAP

WRONG: PV is rapidly fatal and needs urgent chemotherapy at diagnosis. It is typically indolent over decades; the cornerstone is phlebotomy to Hct <45% plus low-dose aspirin, reserving cytoreduction for high-risk (age >60 or prior thrombosis) patients.